CS23007
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REASONING WITH IMPERFECT KNOWLEDGE

Q1:

$P(S1)$ = Prob. that S_1 is late $= 0.7$

$P(S2)$ = " S_2 " $= 0.8$

a) Neither S_1 nor S_2 is late.

$P(\cancel{A \cup B}(S_1 \cup S_2)' = 1 - P(S_1 \cup S_2)$ mutual

$= 1 - [0.7 + 0.8 - 0]$

$= 1 - \cancel{0.75}\ 1.5$

$= 0.05$

For independent events:

$P(A \cap B) = P(A) \times P(B)$

$P(\neg S_1 \cap \neg S_2) = P(\neg S_1) \times P(\neg S_2)$

$= (0.7+1) \times (1 - 0.8)$

$= 0.06$

b) Atleast one of them is late on a given day

$P(S_1 \cup S_2) = P(S_1) + P(S_2) - P(S_1 \cap S_2)$

$= 0.7 + 0.8 - \cancel{0.06}\ (0.7 \times 0.8)$

$= 0.94$

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Age>60

Q2:

$P(Phd) = 0.05$ = Any person chosen at random being PHD

~~$P(Phd / Age>60)$~~ $= 0.045 = P(Age>60 / Phd)$

$P(Age>60) = 0.10$

$P(A/B) = \frac{P(A \cap B)}{P(B)}$

$P(-Phd / Age>60) = ?$

Sol From Cond. prob:

$P(PhD \cap Age>60) = P(Age>60) \times P(PhD / Age>60)$

$= 0.045 \times 0.05$

$= 0.00225$

$P(-PhD \cap Age>60) = P(Age>60) - P(Phd \cap Age>60)$

$= 0.10 - 0.00225$

$= 0.09775$

$P(-PhD / Age>60) = \frac{P(-PhD \cap Age>60)}{P(Age>60)} \Rightarrow \frac{0.09775}{0.1}$

$= 0.9775$

Q1:

a) $P(\text{Heavy Rain}) \Rightarrow 0.05 + 0.1 + 0.1 + 0.05 = 0.30$

b) $P(\text{Storm} / \text{Heavy Rain}) = P(S \cap HR) / P(HR) \Rightarrow \frac{0.05}{0.30} \simeq 0.1667$

c) $P(\text{No Rain} / \text{Strong Wind}) = \frac{P(NR \cap SW)}{P(SW)} = \frac{0.2}{(0.2+0.15+0.1)} = 0.44$

d) $P(\text{no Rain} \cap \text{strong wind}) = 0.2$

Marginal Probability:
Probability of x, no matter what other variables are doing.

Q4:

a) $P(\text{Battery}) = 0.7 + 0.085 + 0.05 + 0.03 \Rightarrow 0.865$

b) $P(\text{Radio}) = 0.7 + 0.05 + 0.002 + 0.03 \Rightarrow 0.0782$

$P(\text{Radio} = \text{not working}) = 0.085 + 0.03 + 0.003 + 0.1 \Rightarrow 0.218.$

We need to find both $\{a, -a\}$ for marginal distributi

a c) $P(\text{Radio} \cap \text{car doesn't start}) = 0.05 + 0.03 \Rightarrow 0.08$

d) $P(\text{radio} = \text{not working} \vee -\text{start}) =$

$0.085 + 0.03 + 0.03 + 0.1 + 0.05 + 0.03$ X

$\Rightarrow P(-\text{radio}) + P(-\text{start}) - P(-\text{radio} \cap -\text{start})$

$\Rightarrow 0.218 + (0.21) - (0.03 + 0.1)$

sum of bottom row

F.

$\Rightarrow 0.298.$

Q5: spam = cause. (features | spam)

a) $P(\text{buy}, \text{discount}, -\text{meeting} \mid \text{spam})$

$P(\text{buy} \mid \text{spam}) \cdot P(\text{discount} \mid \text{spam}) \cdot P(-\text{meeting} \mid \text{spam}) \cdot$

b) $[\because P(\text{effect}_1, \ldots, \text{effect}_n \mid \text{cause})]$

$\Rightarrow 0.5 \times 0.60 \times (1 - 0.10)$

$\Rightarrow 0.27$

- meeting = spam

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b) P(buy, discount, -meeting / -spam)

$\Rightarrow$ P(buy / -spam) . P(discount / -spam) . P(-meeting / -spam)

$\rightarrow$ $0.05 \times 0.03 \times (1-0.4)$

$\Rightarrow$ 0.0009

c) Posterior Probabilities.

$$P(\text{Class} / \text{Features}) = \frac{P(\text{features}/\text{Class}) \cdot P(\text{Class})}{P(\text{features})}$$

P(~~buy/spam~~ spam / buy, discount, meeting)

$$\Rightarrow \frac{P(\text{buy}/\text{spam}) \cdot P(\text{discount}/\text{spam}) \cdot P(\text{meeting}/\text{spam}) \cdot P(\text{spam})}{P(\text{buy}, \text{discount}, \text{meeting})}$$

$\Rightarrow$ $0.5 \times 0.6 \times 0.9 \times 0.4 \times \alpha$ —— (1)

~~0.0027 × α~~ 0.108α

P(-spam / buy, discount, meeting)

$$\Rightarrow \frac{P(\text{buy}/-\text{spam}) \cdot P(\text{disc}/-\text{spam}) \cdot P(\text{meeting}/-\text{spam}) \cdot P(-\text{spam})}{P(\text{buy}, \text{discount}, \text{spam})}$$

$\Rightarrow$ $0.05 \times 0.03 \times 0.96 \times 0.6 \times \alpha$ —— (2)

~~α × 0.00036~~ —— 0.00054α —— (2)

$$\alpha = \frac{1}{0.00054 + 0.27} = \frac{1}{0.27054} \quad \frac{1}{0.108 + 0.0009}$$

Eq (1): P(spam / featu) $\simeq$ 0.995

Eq (2): P(-spam / featu) $\simeq$ ~~0.029~~ 0.005